Attachment 4 — CLI-CPU ↔ Symphact Interaction Diagram

> NLnet NGI Zero Commons Fund — Symphact application, 13th open call (deadline 2026-06-01). Co-design feedback loop between the Symphact actor runtime and the CLI-CPU / CFPU open silicon.

1. The two repositories, deliberately separated

Dimension	`FenySoft/Symphact`	`FenySoft/CLI-CPU`
Deliverable	Software runtime, OS services	Hardware ISA, RTL, silicon tape-out, FPGA
Target	.NET 10 library (Windows / Linux / macOS)	Verilog synthesis, Sky130 PDK
License	Apache-2.0 (permissive)	CERN-OHL-S-2.0 (reciprocal hardware)
Funded milestones	Actor runtime + kernel actors	F2 RTL, F3 Tiny Tapeout, F4 FPGA multi-core
Dependency	None on CLI-CPU silicon (simulator suffices)	None on Symphact

Both projects are developed by the same applicant — **two open projects, one developer, one openly documented co-design loop**.

2. The OS → HW feedback workflow

```
┌─────────────────────────────────────────────────────────────────┐
│  Symphact developer working on an OS milestone                  │
│  (e.g. M0.5 content-addressed journal, M2.5 capability reg, ...)│
└─────────────────────────────────────────────────────────────────┘
                         │
                         ▼  hits a hardware-shaped need
                         │  (profiling, software workaround,
                         │   design wall)
                         ▼
┌─────────────────────────────────────────────────────────────────┐
│  1. Issue opened in FenySoft/Symphact using template            │
│     `.github/ISSUE_TEMPLATE/osreq-for-cfpu.md`                  │
└─────────────────────────────────────────────────────────────────┘
                         │
                         ▼
┌─────────────────────────────────────────────────────────────────┐
│  2. Durable markdown lands in `docs/osreq-to-cfpu/`             │
│     bilingual (en + hu), versioned, status: Draft               │
└─────────────────────────────────────────────────────────────────┘
                         │
                         ▼
┌─────────────────────────────────────────────────────────────────┐
│  3. Linked issue opened in FenySoft/CLI-CPU                     │
│     label: `osreq-from-os`, references Symphact issue           │
└─────────────────────────────────────────────────────────────────┘
                         │
                         ▼
┌─────────────────────────────────────────────────────────────────┐
│  4. CLI-CPU designers consider in the relevant phase            │
│     (F2 RTL, F3 Tiny Tapeout, F4 FPGA, F5 Rich core, F6 silicon)│
└─────────────────────────────────────────────────────────────────┘
                         │
                         ▼
                 Decision = HW implements
                      OR explicit "won't fix"
                      — both are documented under
                      Apache-2.0 / CERN-OHL-S
```

3. Active osreq backlog at submission

Six active OS-requirements landed during the M0.1 → M0.5 work. Each is bilingual, versioned, and linked back to the originating Symphact milestone.

#	Title	CFPU phases	Status	Origin (Symphact milestone)
OSREQ-001	Tree-structured interconnect between cores	F4, F5, F6	Draft	M0.4 scheduler
OSREQ-002	MMIO memory map — OS↔HW register interface	F4, F5, F6	Draft	M0.7 (planned), device actor framework
OSREQ-003	Core reset mechanism — supervisor restart	F4, F5, F6	Draft	M0.3 supervision
OSREQ-004	DMA engine — non-blocking persistence	F5, F6	Draft	M0.5 persistence
OSREQ-005	Mailbox interrupt vs polling — core notification	F4, F5, F6	Draft	M0.4 scheduler
OSREQ-006	Inter-chip link protocol — distributed fabric	F6, F7	Draft	M0.6 remoting (planned)

OSREQ-007 (actor-ref format) was **superseded by the CST model** during the M0.3/M0.4 work — documented as obsolete, with the new CST model integrated into both repos (`docs/trust-model-en.md` v1.1).

4. Expected feedback during the grant (3–7 new osreqs)

Each grant milestone is paired with a **measurement sub-slice** designed to surface concrete HW requirements. Expected feedback paths over the 12 months:

Grant milestone	Expected `osreq-to-cfpu` issue (1–2 each)
M1: Persistence	SHA-256 / BLAKE3 hash instruction (precedent: ARM SHA extensions, Intel SHA-NI); HW mailbox FIFO depth tuning
M2: Remoting + Capability Registry	CST cache or HW lookup-assistance; capability invalidation broadcast primitive
M3: Kernel + Device actors	Supervisor fault-notification line (vs. mailbox watchdog); MMIO mapping discipline
M4: CFPU integration demo	SRAM-to-SRAM DMA for actor-state migration, or an explicit "not needed at this scale" conclusion
M6: Formal-verification foundations	Optional global scheduling clock or sync primitive for verification runs

Total expected: **3–7 concrete `osreq-to-cfpu` issues** across the 12 months — each backed by a benchmark measurement from the simulator (per the discipline rule: *no requirement without data*). Negative results count: a "won't fix in silicon" conclusion is also useful HW guidance.

5. Historical precedents for OS↔HW co-design

The osreq-to-cfpu directory README cites three documented precedents:

Precedent	Year	What was co-designed
Inmos Transputer + Occam	~1984	HW mailbox channels and the Occam `chan` primitive — message passing as an ISA + language feature simultaneously
Symbolics Lisp Machine	1985 (Moon: "Architecture of the Symbolics 3600")	Tagged pointer ISA and GC hardware support, co-designed with the Lisp runtime
Google TPU + TensorFlow	2017 (Jouppi et al., ISCA)	Published co-design where workload requirements measurably shaped chip microarchitecture

Symphact + CFPU adds an **openly documented** fourth entry to this short list: every feedback decision lives under Apache-2.0 / CERN-OHL-S, is reproducible, and is tracked as an `osreq-to-cfpu` issue. The three precedents above show the pattern works (Transputer's HW channels + Occam, Symbolics tagged pointers + Lisp, TPU's matrix unit + TensorFlow); Symphact + CFPU is the modern, fully-open instance.

6. Why this matters for NGI

NGI mission alignment:

- **European paradigm sovereignty** — open licenses, no US/Asian IP dependencies
- **Reproducible engineering** — every feedback decision is documented and verifiable
- **Avoids historical OS/HW mismatches** — x86 segmentation, Itanium VLIW, ARM big.LITTLE rollout, Spectre/Meltdown all originated in HW/OS drift
- **Bidirectional, not waterfall** — the OS does not just "use" the HW; the HW design is shaped by measured OS needs

The same applicant developing both projects means osreq turnaround is **measured in days, not months**, and the next CFPU RTL iteration directly absorbs the OS findings — no separate vendor negotiation, no NDA wall.